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The Impact of Physical Activity and Sport on Overall Mental Health

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Abstract

Background. While physical activity is widely acknowledged for its physiological benefits, its role as a non-pharmacological intervention for mental health stabilization across diverse populations demands reevaluation. Exercise and structured sport participation serve as key systemic regulators of neuroplasticity, mood elevation, and stress resilience.

Aim. This review aimed to synthesize literature concerning the impact of physical activity and sport on depression, anxiety, cognitive function, and psychological well-being, focusing on neurobiological pathways, psychosocial dynamics, and optimal exercise dosing strategies.

Material and methods: Core biomedical and sports science databases were reviewed for cohorts, systematic reviews, and trials investigating exercise modalities (aerobic, resistance, HIIT, team sports), neurobiological biomarkers (BDNF, neurotransmitter regulation), and dose-response dynamics.

Results: Regular physical activity significantly reduces incident depression risk and mitigates anxiety symptoms. Exercise induces structural neural adaptations—including hippocampal volume expansion and BDNF upregulation—while modulating monoaminergic systems and reducing systemic inflammation. Team sports provide superior psychosocial benefits via social integration. Moderate activity (30–60 min/session, 3–5 days/week) yields maximum mental health improvements, whereas excessive volume increases athlete burnout risks.

Conclusions. Physical activity and sport represent high-quality interventions for mental health promotion and affective disorder prevention. Structured exercise acts as a crucial neuroprotective agent that enhances emotional regulation, cognitive resilience, and overall quality of life.

Keywords: mental health, physical activity, sport, depression, anxiety, neuroplasticity, quality of life

Introduction

Mental health disorders, particularly depression and anxiety, represent a major global public health challenge. The growing burden of psychological distress has driven researchers to explore effective non-pharmacological interventions. Among these, physical activity and structured sport participation have emerged as critical modalities for promoting psychological well-being, enhancing mood, and mitigating affective symptoms [3, 19]. While the general health benefits of exercise are well-established, its specific systemic impact on brain structure, emotional regulation, and stress resilience warrants detailed synthesis [14, 15].

Large-scale epidemiological studies demonstrate a clear inverse relationship between physical exercise and mental health burden. Individuals engaging in regular physical activity report significantly fewer days of poor mental health compared to inactive populations [3]. Furthermore, prospective cohort studies confirm that physical activity serves as a protective factor against the onset of incident depression [13, 16] and anxiety disorders [17]. These preventative benefits are observed across diverse demographic groups, highlighting the universal applicability of exercise as a public health tool [15, 18].

The therapeutic efficacy of physical activity operates through interconnected neurobiological, physiological, and psychosocial pathways [9, 10]. At the neurobiological level, exercise stimulates structural and functional neural adaptations. Aerobic training upregulates neurotrophic factors, increases hippocampal volume, and enhances synaptic plasticity [7]. Concurrently, physical exertion modulates central monoaminergic neurotransmission, regulating serotonin, dopamine, and norepinephrine pathways that govern mood and cognitive control [11]. Exercise also dampens systemic inflammatory cascades and lowers chronic physiological stress markers, reinforcing immune and endocrine stability [12].

In addition to physiological responses, the context of exercise plays a fundamental role in shaping psychological outcomes. Participation in team sports provides unique social benefits, fostering peer integration, social identity, and interpersonal support networks [1, 6]. These social dynamics contribute to reduced feelings of isolation and lower rates of anxiety compared to purely individual athletic pursuits [21]. However, the relationship between exercise volume and mental health is not strictly linear. Evidence points to a U-shaped dose-response curve, where moderate, consistent activity yields optimal mental health outcomes, whereas excessive

training loads without adequate recovery can lead to burnout, overtraining, and heightened distress [2, 5].

Given the growing interest in exercise-based mental health interventions, a systematic synthesis of current literature is needed to guide evidence-based recommendations. This paper evaluates the impact of physical activity on mental health by examining its underlying neurobiological mechanisms, psychosocial drivers, and optimal training dosage strategies [4, 8, 20].

2. Research materials and methods

To address the established research problems and provide a high-quality synthesis of the neurobiological, psychosocial, and systemic effects of physical activity on mental health, a methodology for literature searching and data selection was deployed. The research material comprises peer-reviewed primary research articles, randomized controlled trials (RCTs), prospective cohort studies, meta-analyses, and systematic reviews indexed in international scientific databases, including PubMed, Web of Science, Scopus, Google Scholar, and Crossref.

The literature search was executed utilizing combinations of the following standardized medical subject headings (MeSH) and keywords: "physical activity", "exercise", "mental health", "depression", "anxiety", "neuroplasticity", "BDNF", "hippocampus", "team sports", and "burnout". The inclusion criteria limited the core analysis to studies that evaluated:

- The preventative and therapeutic effects of physical exercise on incident depression, anxiety symptoms, and general psychological distress across clinical and non-clinical populations.
- The neurobiological, endocrine, and physiological mechanisms underlying exercise-induced mood elevation, including neurotrophin expression (e.g., Brain-Derived Neurotrophic Factor), hippocampal volume changes, monoaminergic neurotransmission, and inflammatory responses.
- The psychosocial benefits, social integration outcomes, and psychological resilience factors differentiating team sports from individual sports.

- The dose-response relationships, training intensity dynamics (e.g., high-intensity interval training, resistance training), and the psychological impact of overtraining and burnout.

Studies that lacked clear methodological descriptions of exercise parameters, non-peer-reviewed opinion pieces, or trials examining multi-modal therapeutic interventions where the isolated effect of physical activity could not be determined were excluded from the primary synthesis.

2.1. AI

AI was utilized for two specific purposes in this research. Text analysis of clinical reasoning narratives to identify linguistic patterns associated with specific logical fallacies. Assistance in refining the academic English language of the manuscript, ensuring clarity, consistency, and adherence to scientific writing standards. AI were used for additional linguistic refinement of the research manuscript, ensuring proper English grammar, style, and clarity in the presentation of results. It is important to emphasize that all AI tools were used strictly as assistive instruments under human supervision. The AI tools served primarily to enhance efficiency in data processing, pattern recognition, and linguistic refinement, rather than replacing human judgment in the analytical process.

3. Results

The synthesis of the selected literature reveals a multi-faceted framework through which physical activity and sport influence psychological well-being. The findings are categorized into three core domains: neurobiological and physiological adaptation, psychosocial dynamics across sport modalities, and dose-response characteristics alongside overtraining risks.

3.1. Neurobiological and Physiological Mechanisms

The primary biological pathway through which exercise alleviates depressive and anxiety symptoms involves structural and neurochemical remodeling of the central nervous system [9, 10]. Chronic aerobic exercise significantly upregulates Brain-Derived Neurotrophic Factor (BDNF), a primary neurotrophin that stimulates neurogenesis, promotes dendritic branching, and enhances synaptic plasticity, particularly within the dentate gyrus of the hippocampus [7]. Aerobic training over extended periods leads to measurable volume increases in hippocampal

structures, directly mitigating the structural atrophy and loss of neural volume frequently observed in patients with major depressive disorder [7].

Neurobiological / Physiological Pathway	Primary Biological Target	Downstream Psychological and Clinical Outcome
BDNF and Neuroplasticity	Hippocampus (Dentate Gyrus), BDNF gene expression	Structural volume preservation, enhanced cognitive resilience, improved memory function [7]
Monoaminergic Systems	Serotonin, Dopamine, and Norepinephrine receptors	Central mood elevation, heightened motivation, restoration of reward processing pathways [11]
HPA Axis Regulation	Basal Cortisol, Adrenal stress response	Dampened physiological stress reactivity, reduction in chronic autonomic hyperarousal [12]
Inflammatory Cascades	Pro-inflammatory cytokines (IL-6, TNF-alpha)	Reduced neuroinflammation, mitigation of systemic sickness behavior, anxiolytic stabilization [8, 20]

Table 1. Neurobiological and physiological pathways mediating the psychological benefits of physical activity and exercise.

In addition to neuroplastic adaptations, physical exertion exerts a profound influence on central monoaminergic neurotransmission [11]. Acute and sustained exercise increases synaptic concentrations and receptor sensitivity for serotonin, dopamine, and norepinephrine, effectively mimicking the pharmacological mechanism of action of conventional antidepressant medications [11]. Furthermore, physical activity regulates the hypothalamic-pituitary-adrenal (HPA) axis by reducing baseline cortisol secretion and dampening systemic inflammatory cascades characterized by elevated pro-inflammatory cytokines [12]. The

systemic anti-inflammatory effect of regular physical activity contributes directly to anxiolytic responses and long-term mood stabilization across clinical and subclinical populations [8, 20].

3.2. Psychosocial Dynamics and Sport Modalities

Beyond direct physiological mechanisms, the social and psychological context in which exercise takes place exerts a powerful influence on mental health outcomes [10, 14]. Participation in structured physical activity enhances self-efficacy, mastery, and physical self-perception, providing individuals with a heightened sense of personal agency, internal locus of control, and emotional self-regulation [15].

Sport / Exercise Modality	Key Psychosocial Drivers	Primary Psychological Outcomes
Team Sports	Social integration, shared goal pursuit, peer accountability, group identity	Reduced social isolation, lower incidence of anxiety, elevated psychological resilience [1, 6, 21]
Individual Sports	Personal autonomy, self-reliance, intrinsic goal orientation	Enhanced self-discipline, though potentially higher vulnerability to internal stress [16, 21]
Resistance Training	Perception of physical strength, goal attainment, progressive overload	Significant reduction in depressive symptoms across clinical and non-clinical groups [8]
High-Intensity Interval Training (HIIT)	Time efficiency, rapid physiological feedback, acute affective shifts	Improved cognitive control, executive function, and rapid stress clearance [5]

Table 2. Psychosocial drivers and primary mental health outcomes across different sport and exercise modalities.

Comparative analyses between team and individual sports demonstrate distinct psychosocial profiles and differential mental health benefits. Adults and youth engaging in team sports report significantly lower levels of anxiety and depressive symptoms compared to individual sport

participants [1, 6, 16]. Team sports foster social connectedness, peer support, and collective identity, which serve as protective psychological buffers against environmental stressors and feelings of isolation [1, 21]. Conversely, individual athletes often experience higher levels of internal pressure, self-critical evaluation, and performance-related anxiety due to the absence of shared responsibility, requiring targeted psychological support strategies [21].

3.3. Exercise Dosing, Intensity, and Overtraining Risk

The epidemiological data confirm a clear dose-response relationship between physical exercise volume and psychological well-being [3]. The relationship follows a non-linear pattern where optimal mental health benefits are concentrated within moderate activity parameters, specifically 30 to 60 minutes per session performed 3 to 5 times per week [3, 13]. Within this window, physical activity yields a substantial reduction in the overall risk of incident depression [13, 16] and clinical anxiety disorders [17, 18].

Training Volume / Profile	Exercise Characteristics	Associated Psychological State and Risk Level
Sedentary / Inactive	Minimal to no regular physical exertion	High baseline risk for depressive episodes, elevated psychological distress, and low stress tolerance [3, 13]
Optimal Exercise Zone	30–60 min per session, 3–5 days/week (Moderate intensity)	Maximum reduction in depression and anxiety, optimal mood elevation, enhanced life satisfaction [3, 16, 17]
Excessive / Unrecovered	High volume, high frequency, inadequate rest intervals	Heightened risk of overtraining syndrome, elevated baseline cortisol, emotional exhaustion, and burnout [2, 4]

Table 3. Dose-response relationships between physical training volume, psychological state, and mental health risk profiles.

Varying exercise modalities offer complementary benefits depending on training intensity and structure. Resistance exercise training displays a strong, independent antidepressant effect across adult populations regardless of baseline health status [8], whereas High-Intensity

Interval Training (HIIT) yields rapid improvements in cognitive health and executive function in adolescents [5]. Broad overview meta-analyses confirm that structured exercise interventions produce clinically meaningful reductions in depression, anxiety, and general psychological distress across diverse global cohorts [15, 18, 19].

However, exceeding optimal volume thresholds without providing adequate recovery intervals triggers maladaptive psychological and physiological stress responses. Excessive training loads combined with chronic physical fatigue lead to overuse injuries, overtraining syndrome, and athlete burnout [2]. In child and adolescent athletes, early sports specialization and high-volume training correlate with increased psychological distress, elevated baseline stress hormones, loss of intrinsic motivation, and early withdrawal from physical activity [2]. Maintaining an appropriate equilibrium between training intensity, total load, and systemic recovery is therefore essential to preserve the psychological benefits of sport participation [2, 4].

4. Discussion

The findings of this review demonstrate that physical activity and sport participation exert a profound, multi-systemic impact on mental health. The integration of neurobiological mechanisms, psychosocial contexts, and training parameters provides a robust foundation for understanding exercise as both a preventative measure and an effective therapeutic intervention for psychological distress [3, 19].

The neurobiological evidence confirms that physical exercise acts directly on central brain structures to foster psychological resilience [7, 9]. The observed upregulation of BDNF and subsequent preservation or enhancement of hippocampal volume directly counteracts the structural neurodegeneration associated with major depressive disorder and chronic stress [7]. Furthermore, the modulation of monoaminergic neurotransmission (serotonin, dopamine, and norepinephrine) alongside the suppression of pro-inflammatory cytokines explains the immediate and long-term mood-elevating and anxiolytic effects of regular exercise [8, 11, 20]. These biological adaptations closely parallel the mechanisms of conventional pharmacotherapy, reinforcing physical activity as an essential evidence-based adjunct in clinical psychiatry [14, 15].

From a psychosocial perspective, the context in which physical exertion occurs significantly alters its mental health outcomes [10, 14]. While individual exercise modalities such as

resistance training and high-intensity interval training (HIIT) yield substantial cognitive and affective benefits [5, 8], participation in team sports provides distinct advantages [1, 6]. The shared goal pursuit, social support, and sense of community inherent in team environments serve as protective buffers against feelings of loneliness and social isolation [1, 21]. Conversely, individual sport participants face higher risks of internal anxiety and self-critical stress, highlighting the necessity of tailoring psychological support strategies based on the chosen athletic discipline [21].

A critical takeaway from the literature concerns exercise dosing and the non-linear relationship between training volume and psychological well-being [3]. The non-linear dose-response curve indicates that moderate, consistent physical activity (30–60 minutes per session, 3–5 times per week) maximizes mental health benefits [3, 13, 17]. Crossing this threshold without proportional systemic recovery risks triggering overtraining syndrome, emotional exhaustion, and athlete burnout [2, 4]. This risk is particularly pronounced in young athletes subjected to early sports specialization and excessive training volumes, which can convert a healthy psychological outlet into a source of chronic distress [2].

Limitations and Future Research Directions

While the present review synthesizes high-quality evidence across multi-disciplinary domains, several limitations in the current literature must be acknowledged:

1. **Heterogeneity in Methodologies:** Existing studies vary widely in how physical activity is quantified (self-report surveys vs. objective accelerometry) and how psychological outcomes are measured, making direct cross-study comparisons challenging.
2. **Causal Directionality:** Although prospective cohort studies indicate that exercise protects against incident depression [13, 16], reverse causality (where individuals with lower baseline psychological distress are inherently more motivated to exercise) remains a confounding factor in observational designs.
3. **Individual Variability:** Inter-individual differences in genetic predisposition, baseline neuroendocrine profiles, and personality traits influence how a person responds to specific exercise doses and modalities.

Future research should focus on longitudinal randomized controlled trials that combine objective physiological biomarkers (e.g., neuroimaging, systemic inflammatory profiles) with granular tracking of exercise modalities. Furthermore, clear clinical guidelines are needed to define tailored "exercise prescriptions" (modalities, intensities, and rest intervals) optimized for specific mental health conditions.

5. Conclusions

Physical activity and structured sport participation represent high-quality, multi-systemic non-pharmacological interventions for promoting mental health, preventing affective disorders, and supporting psychological well-being across diverse populations. The findings of this synthesis establish that regular exercise acts as a crucial regulator of emotional stability, cognitive function, and systemic stress resilience.

The fundamental conclusions derived from the evidence are as follows:

1. **Neurobiological and Physiological Adaptations:** Regular exercise triggers structural and functional brain remodeling by upregulating key neurotrophic factors (particularly BDNF), promoting hippocampal volume expansion, restoring monoaminergic neurotransmitter balance (serotonin, dopamine, norepinephrine), and dampening systemic neuroinflammatory cascades.
2. **Psychosocial Context and Modality Impact:** While individual exercise modalities such as resistance training and high-intensity interval training significantly enhance self-efficacy, body image, and personal control, team sports offer distinct psychological protection against anxiety and social isolation through shared goal pursuit, peer support, and enhanced social connectedness.
3. **Non-Linear Dose-Response and Recovery Equilibrium:** The psychological benefits of physical activity follow a non-linear pattern. Moderate, regular activity (30 to 60 minutes per session, 3 to 5 days per week) yields maximum mental health improvements and mood elevation, whereas excessive training loads coupled with inadequate physical and mental recovery significantly elevate the risk of overtraining syndrome, emotional exhaustion, and athlete burnout.

In summary, sport transitions from a simple tool for cardiovascular and physical fitness into a vital, multi-dimensional mechanism for mental health stabilization. Implementing exercise parameters that prioritize individual preferences, balanced training loads, and adequate recovery is essential to maximizing psychological resilience while protecting athletes and practitioners from chronic psychological stress.

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Conceptualization: KS; SP: AM

Methodology: AM

Software SP

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Formal analysis: KS

Investigation: KS, AM

Resources: KS; SP

Data curation: KS

Writing- rough preparation: KS

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